Approximation Algorithms

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1 NP-Complete

Techniques to hand NP Compelte

- Brute force (ie exponential time)
- Heuristics
- approximation algorithms
- Fixed parameter tractable algorithms

2 Approximation Algorithms

$ALG \leq \alpha OPT$

- AGL: cost (size) of algorithms solution
- α: approximation ratio
- OPT: cost (size) of optimal solution

If problem is a maximization problem, $ALG \geq \frac{1}{\alpha} OPT$

2.1 Simple Example:

Suppose I know my algorithm is a 1.12-approximation algorithm ($\alpha = 1.12$). Suppose my algorithm returns a solution of cost=746.125. Then, we know that $756.125 \leq 1.12OPT$.

$\implies \frac{746.125}{1.12} = 666.183 \leq OPT \leq 746.125$

The will give us a guaranteed upper and lower bound of where OPT should be. Then, ask yourself if the known solution is good enough for your purposes.

2.2 Vertex Cover

VC: given graph, find smallest subset of vertices such that everye edge in the graph has at least one vertex in the subset.

```
ALG Vertex Cover Pseudo-code:

While uncoverd edge exists
  select both of its vertices
```

Consider a set of edges $E' \subset E$, that do not share vertices. Is there a relationship between the minimum vertex cover and $|E'|$?

$|E'| \leq OPT$, where OPT is the size of actual smallest vertex cover. If we slected fewer than one vertex per edge, we would not have a vertex cover because that edge would not be covered.

Does the size of the algorithm's output relate to a set of edges that do not share verticies? The algorithm finds a specific subset E' since the algorithm is constructed to only pick both of the uncovered edge's vertices. So if we have 10 edges, then the most the algorithm covers is 10*2 (since it selects two vertices for edge edge): $ALG = 2|E'|$.

$$\implies ALG = 2|E'| \leq 2OPT \implies ALG \leq 2OPT$$

To write an approximiation algorithm, the goal is to relate the optimal solution to some structure S, then relating the algorithm to S. We cannot find optimal vertex covers in poly time unless $P = NP$, but this algorithm is at worst 2-times optimal. Two ways to get a better approximation:

1. Find a new algorithm that performs better
2. Analyze the already-existing algorithm and decrease the $ALG \leq OPT$ bounds

2.2.1 Hierarchies:

Show the Computibility Hierarchy

Approximability Hierarchy

- NPO Set
  - APX subset in NPO
    - Vertex Cover is an item in APX
    - PTAS within APX
      - FPTAS subset within PTAS
      - Knapsack problem within FPTAS
      - PO subset within mystery subset (polynomial optimization)
      - shorest path
  - Set cover is an item of NPO (other people proved this, we will trust ti)

APX: optimization problems that can be approximated within a constant ratio

2.3 Set Cover

Set Cover: Given a set of elements (universe), and sets containing those elements, find the smallest number of sets so that every element of the universe is included.

$U = \{1, 4, 7, 8, 10\}$

$S = \{\{1, 7, 8\}, \{1, 4, 7\}, \{8, 4, 7\}, \{4, 8, 10\}\}$

```
Basic Set Cover Greedy Algorithm:

While element of universe is not included
  select S_i with largest number of excluded elements
```

2.3.1 Solving Set Cover

Suppose the universe contains n elements:

$$ALG \leq \alpha OPT$$

- ALG: # of sets selected by the algortihm to cover all n elements
- OPT: # of sets inn an optimal solution cover all n elements

Game plan: bound max number of sets in $ALG\ldots$ by bounding the max number of iterations of the algorithm… by bounding the size of each set added by the algorithm.

For the first set, we want to select the largest set since it covers the most unique elements (since we dont have any elements yet).

$$? \leq |\text{Biggest Set}| \leq ?$$

We are guaranteed to do at least as good as lower bound, guaranteed we don't do better than upper bound. Think about this. In this case, we want to find the relationship between the lower bound. Once we have this relationship, everything else falls into place

Suppose the universe contains n elements. What wcan we say about the first set selected? Answer: It's the biggest!

$$\text{claim: } \frac{n}{OPT} \leq |\text{Biggest Set}|$$

proof:

$$\text{suppose } \frac{n}{OPT} > |\text{Biggest Set}| \leq |Everyset|$$

How many sets does the optimal solution S^{OPT} use: OPT. Then how many elements are covered by those OPT sets?

Suppose the universe contains n elements. The first set selected will have $\geq \frac{n}{OPT}$ elements.

Then, the number of elements remaining after the first iteration is *original # of elements - # elements removed* $\rightarrow n \geq \frac{n}{OPT}$

$$n_1 \leq n - \frac{n}{OPT} = n\left(1 - \frac{1}{OPT}\right)$$

What can we sayabout the second set selected? **It covers the most uncovered elements**.

- Suppose the first set *was* in the optimal solution
  - Then, a remaining set must have at least $\frac{n_1}{OPT-1}$ uncovered elements.

$$\implies \frac{n_1}{OPT} < \frac{n_1}{OPT-1} \leq |SecondSet|$$

- Suppose the first set *was not* in the optimal solution
  - If not, how do the remaining $OPT - 1$ optimal sets cover the remaining n_1 elements?

$$\implies \frac{n_1}{OPT} \leq |SecondSet|$$

In general, after t iteration:

$$n_t \leq n_{t-1} - \frac{n_{t-1}}{OPT} = n_{t-1}\left(1 - \frac{1}{OPT}\right) \leq \ldots \leq n\left(1 - \frac{1}{OPT}\right)^t$$

The question is what is t? When we solve t, we solve ALG.

Big picture:

1. How many sets are added each iteration: 1
2. $ALG = \#$ of iterations
3. When does the algorithm terminate: $n_t < 1$

What value of t makes $n_t < 1$:

$$\implies ALG \leq ln(n)OPT$$

It turns out that Set Cover cannot be approximated within the bound of $(1 - o(1))ln(n)$, unless P = NP

3 Knapsack problem

Knapsack: given a set of n items with values $v_1, \ldots, v_n$ and weights $w_1, \ldots, w_n$, select the most valuable combination with total weight $\leq W$

3.1 DP method

How could we approach this? **Dynamic programming sense?**

Algorithm: is there an optimal substructure? I.E., if I have an optimal solution to some instance, does that imply an optimal solution to a different instance?

***He's going to talk about the intuitive DP version of the problem, then explain why it wouldn't work. ***

$$\text{Define } opt(i, w) = max\,(opt(i-1, w), opt(i-1, w-w_i) + v_1)$$

$$E[X_e] = 1(2/3) + 0()1/3 = 2/3$$

What now:

- for every single weight $\leq$ W and i, calculate $opt(i, w)$
- no longer polynomial
- reducing search space of weights will compromise weight accuracy. not allowed!

If you make a table with items as the rows and total weights as the columns for the DP, how can you check every possible weight? Its trying to turn something continuous into something discrete, and adding different increments of weights will be difficult and it breaks the DP.

Instead, we need to compromise value accuracy isntead of weights.

3.1.1 Values instead of weights

Instead of saying "for a given weight, how much value can we get?" to "given a value threshold, whats the smallest weight i can get?"

Now we have a table with items as the rows and total value as the columns. For every possible total backpack value, go to the bottom row and check each column until the weight threshold is crossed, then we just go back one spot. That will lead to the optimal solution.

Define $opt(i, V)$ = minimum weight of subset of items $\{1, \ldots, i\}$ that gives value at least V (target value).

Case 1: if i is not in the optimal solution for $\{1, \ldots, i\}$:

- $opt(i, V) = opt(i-1, V)$

Case 2: if i is in the optimal solution for $\{1, \ldots, i\}$:

- $opt(i, V) = opt(i-1, V - v_i) + w_i$

We are going to assume the values are integers

Algorithim:

1. For $0 \leq i \leq n$ and $0 \leq V \leq n$ max(i under max idk how to type that) v_i, compute $opt(i, V)$.
2. Return $V* = max\{V : opt(i, V) \leq W\}$

Running time: $O(n^2 \text{max (i under max)} v_i)$

To make the running time better, we can change the units of the values of the table to lower values, which will increase, but it's not easy for compliated numbers and you can lose information. See below, how 500000 keeps all the information, but the 163882 loses 882 dollors of accuracy:

$$500000 \text{ dollars } \rightarrow 5 \text{ hundred-thousand dollars}$$

$$163882 \text{ dollars } \rightarrow 160 \text{ hundred-thousand dollars}$$

Algorithm:

1. Let $v_{max} = \text{max i under max} v_i$
2. For each i, let $v_i' = \lfloor v_i \frac{n}{\epsilon v_{max}} \rfloor$
3. Run dynamic programming algorithm using $\{v_i'\}, \{w_i\}, W$

Running Time $= O\left(\frac{n^3}{\epsilon}\right)$

How does scaling values impact the cost? (how does ALG relate to OPT)

$\$\mathrm{S}_{\mathrm{ALG}}$ = set the algorithm selected. $\$\mathrm{S}_{\mathrm{OPT}}$ = set the optimal items.

Need to relate $\sum_{i \in S_{OPT}} v_i$ relate to $\sum_{i \in S_{ALG}} v_i$

But the algorithm operates on v_i':

Need to relate $\sum_{i \in S_{OPT}} v_i$ to $\sum_{i \in S_{OPT}} v_i'$ to $\sum_{i \in S_{ALG}} v_i'$ to $\sum_{i \in S_{ALG}} v_i$

$\sum_{i \in S_{OPT}} v_i' = \lfloor v_i \frac{n}{\epsilon v_{max}} \rfloor$

He does algebra:

$\sum_{i \in S_{OPT}} v_i' \leq OPT \frac{n}{\epsilon v_{max}} - n$

How does $ALG = \sum_{i \in S_{ALG}} v_i$ to $OPT = \sum_{i \in S_{OPT}} v_i$?

He does algebra:

$\sum_{i \in S_{OPT}} v_i' \leq OPT \frac{n}{\epsilon v_{max}} - n$

ALG is optimal on the scaled values: ALG is the beest at identifying the **highest *scaled* amount** of value while OPT is the best at identifying the **highest amount** of value in the backpack.

We test OPT on how well it can perform on the scaled values, and it performs worse than ALG because it's playing a game it wasn't designed to do; OPT is not OPT on the scaled values, but ALG *is*.

Im not writing all this math. look at the slides if you want. (end of lecture 4/16).

We relate OPT to ϵv_{max}.

Performance guaranteee: $ALG \geq (1 - \epsilon) OPT$

Running time: $O\left(\frac{n^3}{\epsilon}\right)$

This is called a fully polynomial-time approximation scheme (FPTAS)

We can solve Knapsack instances arbitrarily close to optimal in polynomial time!

4 Randomization

4.1 Graph Coloring

Family of graph problems that involve coloring vertices (or edges) subject to various constraints.

Minimum Vertex Coloring: Color the vertices of a graph useing the smallest number of colors so adjacent vertices are diffreent colors. This is NP-hard.

4.1.1 Max 3-Coloring

An edge in a graph is *satisfied* if its nodes are colored different colors. Using only three colors, color the nodes of a graph such that the number of satisfied edges is maximized.

Algorithm: For each vertex, randomy assign it one of the three colors with a $\frac{1}{3}$ probability.

- Running time $O(n)$
- Valid? Yes, colors all nodes with ≤ 3 colors.
- Optimal? probably not...

Note: If problem is a max problem: $ALG \geq \frac{1}{\alpha} OPT$
Goal: show this algorithm is *expected to perform* as a $\frac{3}{2}$ -approximation algorithm.
What is the probability that a single edge is statisfied?

- All possibilities: (bb, bg, br, gb, gg, gr, rb, rg, rr)
- $\frac{6}{9} = \frac{2}{3}$

Define $X_e = 1$, if e is satisfied, 0 if e is not satisfied
What is $\sum_e X_e$?

- ALG - It's the number of satisfied edges.

How many edges we expect to satisfiy, and we can relate this to OPT (Since $OPT \leq |E|$ (cant have more satisfied edges than there are edges!)):

$$E[ALG] = E[\sum_e X_e] = \sum_e E[X_e] = \sum_e \frac{2}{3} = \frac{2}{3}|E|$$

$$\implies E[ALG] = \frac{2}{3}|E| \geq \frac{2}{3} OPT$$

4.2 SAT & 3-SAT

$$\phi = (x_1 \vee x_1 \vee x_2) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_2}) \wedge (\overline{x_1} \vee x_1 \vee x_2)$$

ϕ is a formula with clauses composed of Boolean variables connected by ORs, and clauses connected by ANDs. Can you set the variables to **true** or **false** so taht ϕ evaluates to **true**?

$$\phi = (x_1 \vee x_1 \vee x_2) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_2}) \wedge (\overline{x_1} \vee x_1 \vee x_2)$$

Solution:$x_1 = \textit{false}$, $x_2 = \textit{true}$

$$(F \vee F \vee T) \wedge (T \vee F \vee F) \wedge (T \vee T \vee T)$$

$$T \wedge T \wedge T \implies T$$

SAT = {<ϕ>: ϕ is a satisfieable formula }
3-SAT = {<ϕ>: ϕ is a satisfieable formula with 3 variables per clause}

4.3 Max 3-SAT

Max 3-SAT Problem: Given a 3-SAT instance (with unique variables in each clause), set variable values so that the number of satisfied clauses is maximized.

Algorithm: Assign x_i = true with probabilities 1/2. Otherwise, set it to false.

We will follow the same technique that we used for max 3-color.

Probabilities that a clause is satisfied = $\frac{7}{8}$.

Define $\mathrm{Z_c} = 1$ if clause c is satisfied, 0 otherwise.

$$E[Z_c] = 1\left(\frac{7}{8}\right) + 0\left(\frac{1}{8}\right) = \frac{7}{8}$$

Number of clauses we expect:

$$E[ALG] = E[\sum_c Z_c] = \sum_c E[Z_c] = \sum_c \frac{7}{8} = \frac{7}{8}|C|$$

$$\implies [ALG] = \frac{7}{8}|C| \geq \frac{7}{8}OPT$$

4.4 ILP-Based Approximation

4.4.1 Vertex Cover ILP

He defines VC again, then shows the ILP for it. We can change the ILP (NP) to an LP (P), then say that our P solution isn't that far off from the NP solution. He has the ... **Check lecture for this part**

What is the relationship between ALG and OPT?

Can we bound OPT from below?

- Let $\mathrm{x_{ILP}}$ and $\mathrm{x_{LP}}$ be the set of x values found by the ILP and LP

Claim (proved): $\sum x_{LP}\ leqOPT$

How does $\sum \mathrm{x_{LP}}$ relate to ALG?

$\sum x_{LP} = \sum_{x_i \in x_{LP}} x_i \geq \sum_{x_i \in x_{LP}: x_i \geq \frac{1}{2}} x_i$ beacuse it's a subset of x_{LP}.

$\geq \sum_{x_i \in x_{LP}: x_i \geq \frac{1}{2}} \frac{1}{2}$, because each x_i is at least $\frac{1}{2}$.

equation i didnt have time to write here

What is the relationship between ALG and OPT?

$\sum x_{LP} \geq \frac{1}{2}ALG$ and $\sum x_{LP} \leq OPT$

What is preventing us from making the 1/2 fraction higher?

5 Randomized Rounding

5.1 Set Cover

If you don't know the goal, look it up.

We are gonna build an ILP for it, relax it to an LP, then do some magic.

ILP: (math notation on lecture slides)

- Decision Variables: Whether we select a set or not
- Objective: minimize number of sets selected to cover entire universe
- Constraints: every single element of the universe is in at least one selected set.

WThis is an ILP because you cannot select portions of sets

We are going to relax it, so $0 \leq x_s \leq 1$ for each set s, then if $x_s \geq 1, addsetstooutsubset$

Why does this work for vertex cover and not set cover? Come back to the lecture on this.

The LP will give us numbres for x_s, then we are using those numbers as our probability. Instead, we are going to add set s to our subset S_{ALG} with probability of x_s. Basically, whats happening is we are using the relaxed ILP (LP) to give us the continuous values for x_s, then we are using those x_s's (there is one per set), choose a random set, then choose to add it with the probability x_s.

Concerns about this aglorithm:

- validity isn't guaranteed, but we can run the algorithm again for a chance for better validity. However, this can change running time since we don't know how many times we need to run the algorithm for the output to become valid.

We need to pick where the uncertainty lies (validity, the second one, running time) and try to account for it.

---

1. What is the size of the solution?
   - Let x_s^* be the optimal solutions to the LP relaxation. These are the probabilities we get from the LP output.
   - Define random variable $X_s = 1$ if s is in alg, 0 otherwise
   - To relate ALG to OPT, we will use the expected value. ALG is the sum of X variables set to 1 that came from the LP.

$$E[\sum_{s \in s_{ALG}} X_s]$$

The expected value of x_s is the probability of x_s being 1, which is the value from the linear program. Which means the Expected value of our ALG is the sum of x_s^*, since x_s^* is the probability of $X_s = 1$

This is the objective function for the LP: $\sum_{s \in S} x_s^* = OPT_{LP} \leq OPT$.

This is saying that out Epected value of ALG is *better than or equal to* OPT. What does this mean? Well, we can't do better than OPT because it's OPT... So this means our validity is probably really bad.

2(two). What is the probability solution is valid?

- Let S_u be the sets of S that contain element u
  - Let S_u be sets of S that contain element u. u is covered by $S_{ALG} \iff \sum_{S_u} X_s \geq 1$.

$P[u \text{ not covered by } S_{ALG}] = \prod_{s_u}(1 - x_s^*)$, since X_s independent

He then talks about how we can bound the product with algebra. For now, just agree that:

$$P[u \text{ not covered by } S_{ALG}] = e^{-\sum_{s_u} x_s^*} \leq \frac{1}{e} \approx 0.37, \text{ since } \sum_s {}_u x_s^* \geq 1$$

What did we accomplish? Its highly unlikely S_ALG will cover all U, but each individual element has good likelihood of being covered. We can run it, get some sets we picked, see it's not valid, keep some of the sets, run the algorithm again with some of our kept sets, and see if any sets got added. Basically, instead of starting over, we grow our collection of sets every time we run it instead of throwing possible solutions out every time we run the algorithm.

—

Algorithm: We will solve the LP, then perform the following:

1. Add set s to out subset $\{S_{ALG}\}$ with probability of x_s.
2. Repeat step 1 T-times, while adding sets to S_{ALG}.

—

Supposed $T = ln4n$, where $|U| = n$ (ln4n isn't picked randomly, it makes the math work nicely.)

1. What is the size of the solution?

Expected value of ALG is the sum of x_s.

$$E[ALG] = T\sum_{s\in S} X_s^*, \text{ by previous bound}$$

What we want to show: $ALG \leq 4ln(4n)OPT$ We will show this by showing the chance that $ALG \leq 4ln(4n)OPT$ does *not* happen (i.e., when will it fail?), aka what is the probability that this fails?

$$ALG \leq 4ln(4n)OPT \leq P[ALG > 4T\sum_{s\in S} x_s^*]$$

The equation above is a general statement that the probabilty of some event increases when the threshold decreases. Like changing "whats the probability of your child being taller than 6 feet" to "whats the probability of your child being taller than 4 feet".

$$\text{Markov's inequality: } P[X \geq a] \leq \frac{E[X]}{a}$$

When you use Markov's inequality on our inequality, we get $\frac{1}{4}$. Look at lecture if you want to show this.

—

2(two). What is the probability solution is valid?

- He shows the probabilty of a single element not being covered is $\frac{1}{4n}$ in the lecture.

other things happen.

—-

The solution is valid with a probability of 3/4, and the probability of the running time(check if this is true) , the probability is greater than or equal to $\frac{9}{16} \approx .56$..

If we run this n times: the probability of the run being good is $\left(1 - \frac{7}{16}\right)^n$

The more we run it, the more likely the algorithm will work. This is a tradeoff.

We have a high likelihood of a good solution with guaranteed polynomial run time.

Can put our uncertainty onto running time instead of "goodness"? Yes!.